

# Design and Implementation of IoT-enabled Compost Monitoring System

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**Abstract**—Precision agriculture has seen growing adoption by farmers as it offers great benefits in terms of productivity, traceability and efficient operation. The areas of application include crop management, livestock management as well as other agricultural product management. This paper presents the development and implementation of compost monitoring system that is enabled by Internet-of-Things (IoT) technologies. This system is designed to be deployed in an organic farm in Serian district, Sarawak, Malaysia. The design of the system setup considers the field conditions at the farm. The two-level (sensor node and gateway) approach in developing the system, combined with the use of LoRa communication and MiFi device permits users to monitor the compost parameters remotely. The experimental results are presented.

**Keywords**—Compost Monitoring, Wireless Sensor Node, Internet-of-Things, LoRa Communication

## I. INTRODUCTION

Organic agriculture holistically entails the practice of building a balanced soil quality and water quality management while conserving biodiversity to maintain healthy habitat patterns. By advocating its stance in preserving the ecosystem (despite the misconception to be ideologically-driven) along with the pressing need for a prudent management of natural resources, this paper intends to contribute in finding an approach to embrace technology in agriculture driving it towards precision farming.

At the very core element of agriculture, soil quality is essential for a sustainable organic farming. To resolve the problem of severe soil degradation due to intensive chemical-laden farming methods, compost is by far one of the best options for soil amendment [1] to replenish organic matter and nutrients. Hence, this research focuses on real-time monitoring of the fundamental parameters of the soil to improve composting efficiency using the application of Internet-of-Things (IoT).

Effective composting method is essential to produce good quality compost and to improve production rate, making it a process-oriented approach. The art and science of composting involves interdependent bioprocesses from complex symbiotic interactions between microbes, to the aerobic conditions down to the particle size. For a successful decomposition,

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manipulation of biological process is required to obtain the ideal conditions for the metabolic processes of the microbes.

Proper composting will follow through four phases: (A) Mesophilic, (B) Thermophilic, (C) Cooling and (D) Compost Maturation [2]. Different population of microorganisms thrives in different temperature ranges and it should occur in phases A and B of the process. For aerobic composition to occur, optimum temperature to stimulate Mesophilic organism growth is between 25 – 40°C while the favourable condition for Thermophilic organism is within the range of 40 – 70°C [3].

Thus, the work aims to seek ways to improve the success rate of the composting process by monitoring the temperature to assist in any manual intervention when required. The proposed automated monitoring system is built with the help of sensors and wireless network to transmit live data of the compost from the field to the farm monitoring system on the cloud server. This allows for off-site remote monitoring of the compost thereby reducing manual work. By combining the findings with the technologies, it will provide a valuable insight to the understanding of the soil biological activity and improve the effectiveness of aerobic decomposition at a finer level. This will contribute in shaping the local scene to a more sustainable organic farming.

## II. METHODOLOGY

### A. Design Considerations and Proposed System Setup

The compost monitoring system was implemented in Satoyama Farm 52, Serian, Sarawak, that produces compost and other organic agricultural products and poultry. The composting works were done in a Compost House that is about 500 meters away from the farm office. The considerations of setting up the monitoring system were determined by the compost parameters, the field conditions and the terrain of the farm.

In terms of compost parameters, the temperatures at different points of the compost pile are essential as they reflect different phases of the composting [4] as elaborated in the previous section. The ambient temperature and relative humidity around the vicinity of the compost are also monitored as they may affect the fluctuations of the compost temperature.

In regards to the accessibility of the data captured, it is expected that the compost information is accessible via the

Internet. This poses a challenge due to the absence of Internet connectivity inside the farm.

Taking into consideration the information discussed in the previous subsection, this work proposes a two-level approach for the compost monitoring. First, a sensor node that carries all the necessary sensors were assembled and deployed at the Compost House. The sensor data is then transmitted to a gateway located in the farm office.

Due to the non-flat terrain and tree obstructions between the farm office and the compost house, it is impractical to install network cables for the transmission of data. In addition to that, in order to facilitate future system scalability, it is only sensible that the communication between the sensor node and the gateway be made by wireless means. With its low power consumption and long distance capabilities, LoRa wireless communication [5] was selected for the transmission of sensor data to the gateway.

The gateway is connected to the Internet through Mobile Internet provided by a MiFi modem to upload the data captured from the node to the cloud storage.

### B. Hardware Selection

The proposed system setup is developed as a wireless sensor node and an IoT gateway, each comprising its own electronic hardware components. Fig. 1 shows the electronic system of the wireless sensor node.

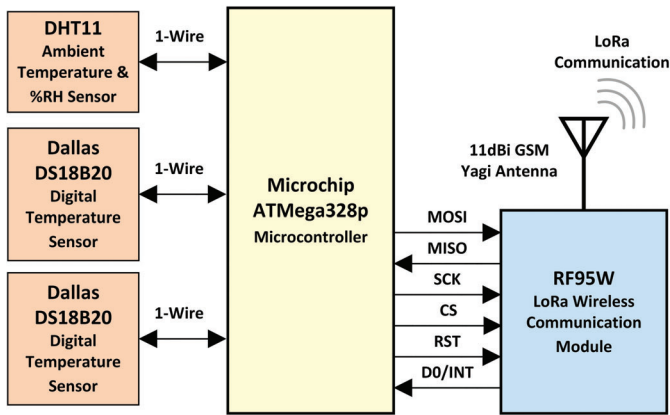


Fig. 1. Block diagram of the wireless sensor node electronic system

The microcontroller, sensors and communication devices used in the wireless sensor node and their functionalities are summarized as follows:

- Microchip ATmega328p is an 8-bit microcontroller. This microcontroller handles the interfaces of the sensors that monitor the compost pile variables as well as the wireless communication module.
- 1 x OSEPP Electronics DHT11 ambient temperature and relative humidity sensor. This sensor collects the ambient temperature and relative humidity near the vicinity of the monitored compost pile. This is interfaced with the microcontroller using 1-Wire interface.
- 2 x Dallas DS18B20 digital temperature sensor with waterproof probe enclosure. This sensor measures

compost temperature in °C and is interfaced with the microcontroller using 1-Wire interface.

- RFM95W LoRa wireless module. This module is a LoRa communication transceiver. This works uses a 919MHz band. The microcontroller communicates with this device via SPI interface.
- RF Solutions 11dBi GSM Yagi Antenna is connected to the LoRa wireless module to transmit sensor information to the LoRa-enabled IoT gateway.

Fig. 2 shows the block diagram of the LoRa-enabled IoT gateway. The summary of the components used in the IoT gateway are as follows:

- Raspberry Pi 3 B single-board computer. This processor handles interfaces of the LoRa wireless module and sends the received sensor data to cloud storage through Internet connectivity.
- USB-powered 4G MiFi Modem. This device provides broadband Internet connectivity required by the single-board computer to upload sensor data to cloud storage through HTTP methods.
- RFM95W LoRa wireless module. The Raspberry Pi single-board computer communicates with this device via SPI interface.
- Siretta 5dB MIKE 2A omnidirectional antenna is connected to the LoRa wireless module to receive sensor information send by the wireless sensor node.

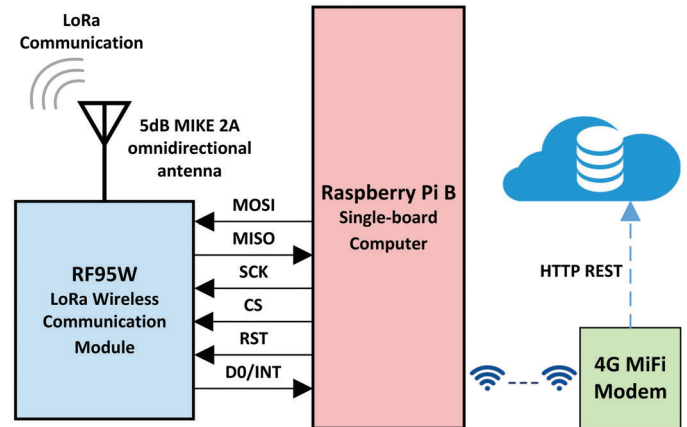


Fig. 2. Block diagram of LoRa-enabled IoT gateway electronic system

In both the node controller and IoT gateway computer, the LoRa modules are interfaced through the standard SPI communication where LoRa modules act as Slave devices and the node controller and gateway computers act as Master devices. Besides the standard SPI lines (MOSI, MISO, SCK and CS), the controllers are able to supply RST signal to reset the LoRa modules. The D0/INT line is a data output from the LoRa module that signifies an incoming data is received by the module and ready to be read by the Master device. This line is connected to an input interrupt pin of the gateway computer. In this case, upon an interrupt request, the gateway computer is able to read the available incoming data.

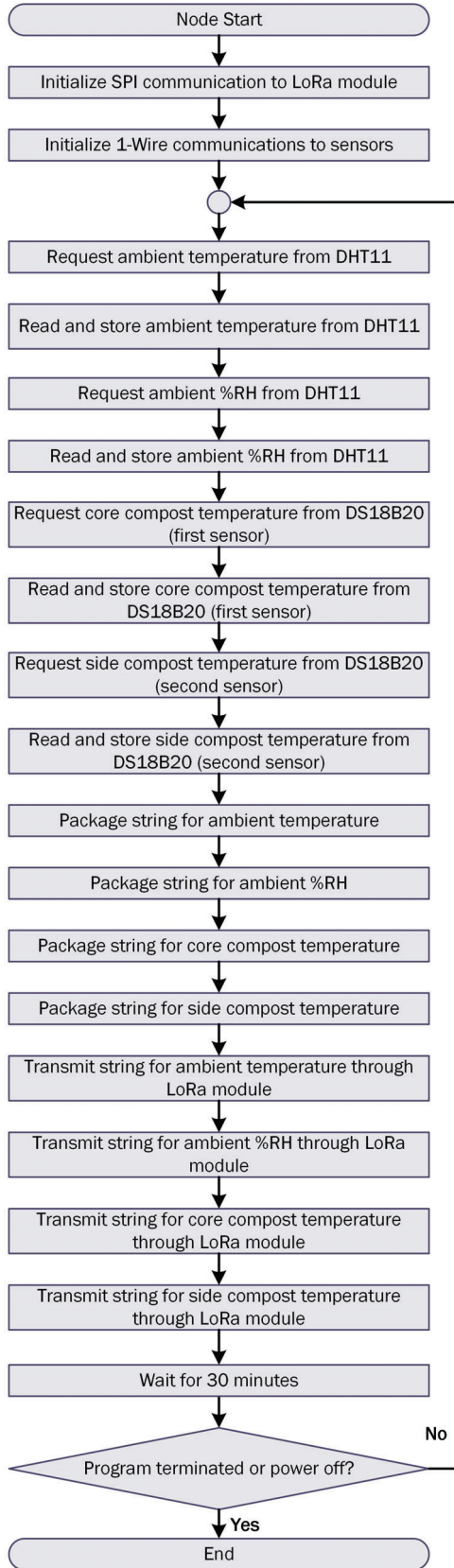


Fig. 3. Flowchart of the code implementation for wireless sensor node microcontroller

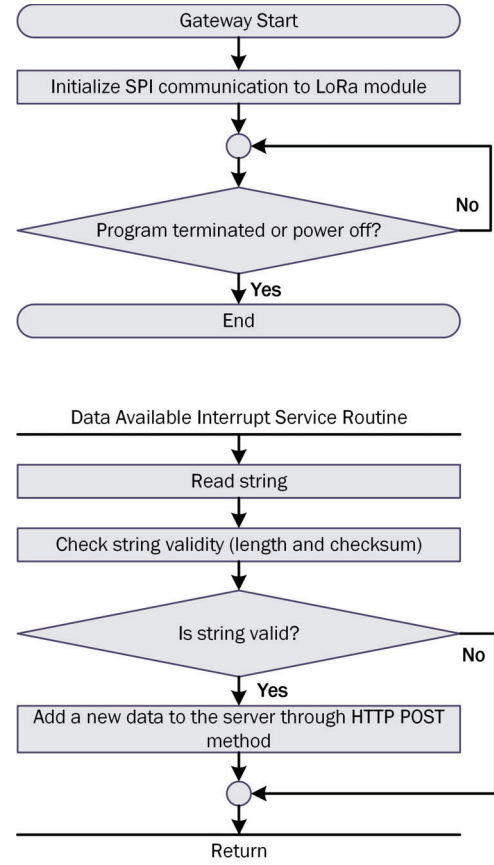


Fig. 4. Flowchart of the code implementation for IoT gateway

### C. Data Collection and Process Flow

The software of compost monitoring system is implemented in the microcontroller for the wireless sensor node and in the Raspberry Pi IoT gateway.

Fig. 3 shows the code implementation of the node microcontroller. Upon starting, the SPI communication between the microcontroller and LoRa module is established. Following that, the 1-Wire connections with the sensors are initialized. The microcontroller then requests, reads and stores the ambient temperature, ambient relative humidity ((core and side). Each of this information is then package to a fixed-length string with a checksum character before it is transmitted through the LoRa module in sequence. The sampling and data transmission is repeated every 30 minutes as long as the node controller is powered and not in reset.

Fig. 4 shows the code implementation in the gateway computer. Upon the running of the program, the SPI communication to the LoRa receiving module is initialized. By virtue of the Data Available trigger signal given by the LoRa module, upon an interrupt request to the gateway computer, the Data Available Interrupt Service Routine (ISR) is handled. The ISR read the incoming string and check whether the string is valid by verifying the string length and checksum character. If the string is valid, the string is uploaded to the cloud storage via HTTP REST POST method. Regardless of



Fig. 5. Deployed wireless sensor node on a freshly-built compost pile

the string validity, the ISR will return to the main routine.

### III. SYSTEM DEPLOYMENT

The wireless sensor node described in Section II was deployed in the farm compost house as shown in Figs. 5 and 6. The sensors were protected by enclosures and placed in the compost pile. The node circuitry is implemented in a weatherproof metal enclosure with the Yagi antenna installed facing the farm office. The IoT gateway comprising Raspberry Pi computer, LoRa module and MiFi modem was deployed in the farm office (Fig. 7) with the omnidirectional antenna placed outside the office (Fig. 8).

The choice of using Yagi antenna at the transmitting end and omnidirectional antenna at the receiving end allows reliable data transfer and this setup will be very suitable for future extension of this work due to the fact the communication system of the additional wireless sensor nodes can be implemented in similar manner as the compost sensor node while maintaining a single IoT gateway.

### IV. RESULTS AND DISCUSSION

Fig. 9 shows data collected by the compost monitoring system over the span of three weeks from the day the compost was built. The average data for each day is summarized into four data points in a quarter-day (6 hours) interval. The first interval (first data point) of the day is from 12midnight-6am, second interval 6am-12noon, third interval 12noon-6pm and fourth interval 6pm-12midnight.



Fig. 6. Transmission of sensor data via LoRa communication through 900/1800MHz Yagi Antenna



Fig. 7. Deployed IoT gateway in farm office



Fig. 8. Receiving of sensor data via LoRa communication through 900/1800MHz omnidirectional antenna

The four sets of data collected are the Ambient Temperature, Ambient Relative Humidity, Compost Temperature 1 (core temperature) and Compost Temperature 2 (side temperature).

The ambient temperature and relative humidity data show typical patterns of the day. The ambient temperature reaches its maximum during 12noon-6pm interval. At this interval, the ambient relative humidity reaches its minimum. During the 12midnight-6am interval, the ambient temperature reaches its minimum and the ambient relative humidity reaches its maximum. This pattern is very significant on normal days, except for days with continuous rain (Days 3, 4, 8, 11 and 12).

The compost temperatures reached their peaks in the first four days when the compost progressed from fresh pile to Mesophilic and Thermophilic phases. As the bacteria activities were taking place very rapidly, the compost temperatures rise. The core temperature (Compost Temperature 1) reached a peak of around  $5^{\circ}\text{C}$  and the side temperature (Compost Temperature 2) reached a peak of around  $43^{\circ}\text{C}$ . As the decomposing progressed, the compost finally settles near the ambient temperature. The compost temperatures, however, were slightly influenced by the ambient temperature. There is about  $1 - 2^{\circ}\text{C}$  increase in temperature at the 12noon-6pm intervals of the day.

The composting phases (A) Mesophilic, (B) Thermophilic, (C) Cooling and (D) Compost Maturation over time can be identified from visual inspections of the changes in Compost Temperatures 1 and 2. Note that the demarcations of the composting phases were approximate.

## V. CONCLUSION AND FUTURE IMPROVEMENT

In this work, the design and implementation of an IoT-enabled compost monitoring system have been discussed.

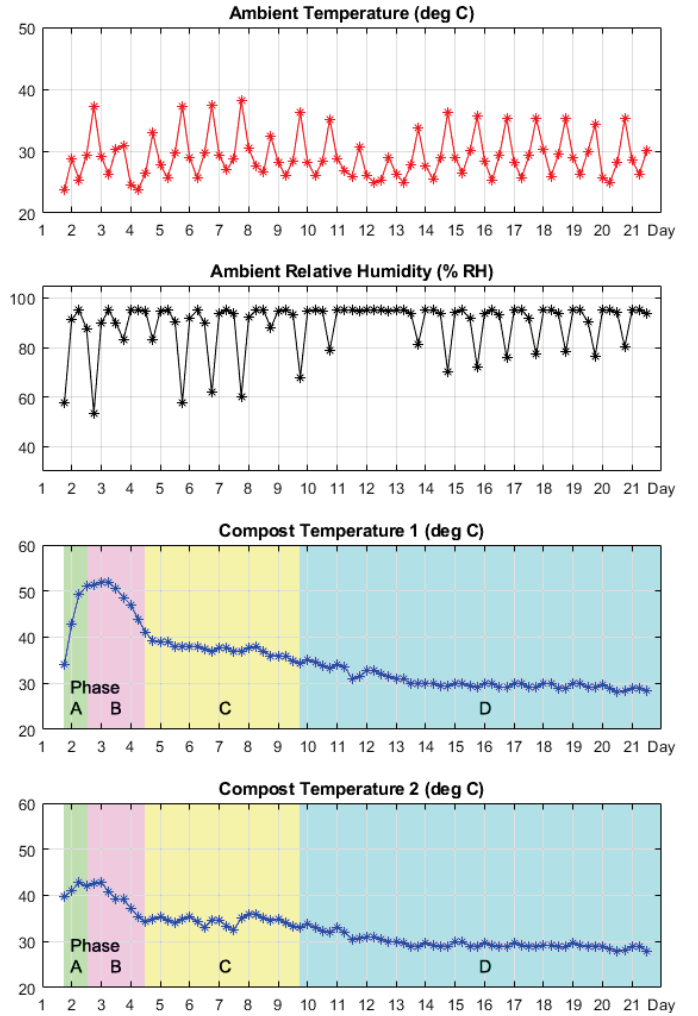


Fig. 9. Collected sensor data for the span of three weeks

The system and the hardware setup chosen in this work provide the agriculture practitioners the convenience in monitoring the compost information remotely and to decide the steps taken within the ongoing composting process.

The hardware setup proposed in this work is suitable for managing farms with diverse ecosystem, where different types of crops, livestock and composting houses are to be precisely monitored over large area. For example, each crop may be monitored by a wireless sensor nodes and the data from multiple nodes are then transmitted to a single IoT gateway that sends the information to the cloud storage.

This work carries the potential to be extended from monitoring and data collection to smart data processing and integrated decision support for the agriculture practitioners. For example, abrupt changes in the ambient and compost temperatures detected from the sensors may be processed to alert the farmers to attend to the compost immediately. Such system may also be used to automatically notify the farmers during the composting period to help the farmers to decide the most suitable time to turn the compost.

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